Ali Abutaleb, PhD

Senior GIS Specialist | Enterprise GIS and Spatial Data Management | Infrastructure and Engineering Projects

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PROFESSIONAL SUMMARY

Saudi GIS specialist and engineer with 20 years of experience across government infrastructure, industrial city development, and heavy industrial engineering projects. Designs and maintains enterprise GIS databases (ArcGIS Enterprise, ArcGIS Online, PostgreSQL/PostGIS, ArcSDE), enforces project data models and national standards (GEOSA, ISO 19115, OGC), and integrates GIS with BIM/CAD models, drone surveys, LiDAR, GPS field data, and satellite imagery. Currently delivering GIS on two concurrent Saudi government programs: the Ministry of Hajj and Umrah GIS platform covering infrastructure across Makkah, Mina, and Arafat, and geospatial studies for the Royal Commission for Jubail and Yanbu (RCJY) at Jazan City for Primary and Downstream Industries (JCPDI). Earlier career includes landslide hazard consulting for the City of Los Angeles and six years as a process engineer on cement plant construction and expansion projects. PhD in geoinformation sciences and author of a BIM-in-ArcGIS cookbook. Supports engineering and design teams with decision-ready maps, advanced spatial analysis, and Python automation.

CORE COMPETENCIES

- Enterprise GIS Database Design and Administration
- Geospatial Data Management and Governance
- Data Models, Schemas, and Standards Compliance
- BIM/CAD and GIS Integration
- Advanced Spatial Analysis and Modeling
- Engineering and Design Team Support
- Infrastructure and Asset Mapping
- Metadata Management and QA/QC
- Python Geoprocessing Automation
- Drone, LiDAR, and Remote Sensing Data
- Web GIS, Dashboards, and Field Apps
- Stakeholder Training

PROFESSIONAL EXPERIENCE

GIS Systems Consultant **Jan 2025 – Present**

Ministry of Hajj and Umrah | Makkah, Saudi Arabia

- Manage GIS data for Hajj infrastructure across Makkah, Mina, and Arafat: service points, crowd control routing, emergency response zones, and readiness layers serving 2+ million pilgrims.
- Administer the Ministry's enterprise ArcGIS Online organization of 166 hosted feature layers; built a Python (ArcGIS API for Python) backup system covering the full organization.
- Led a metadata standards audit that found **83% of layers non-compliant**, then built automated remediation scripts that brought the catalog to **100% compliance**.
- Built and maintain an enterprise PostGIS/ArcSDE geodatabase for infrastructure assets, routing networks, and service points; defined data models and specifications in workshops with senior Ministry stakeholders.
- Integrated drone imagery, satellite imagery, and GPS field surveys into one platform; deployed ArcGIS Field Maps, Survey123, and Dashboards for real-time field data collection and operational monitoring.
- Automated geoprocessing, incident reporting, and routing analysis with Python (ArcPy); performed root-cause analysis of ArcGIS Online credit consumption to control platform operating costs.
- Produced a comparative spatial analysis of the 1446 and 1447 AH Hajj seasons, delivered as Arabic reports, Excel workbooks, and HTML visual reports for senior leadership.
- Trained Ministry teams on platform tools and field data collection protocols; ensured compliance with GEOSA national spatial standards and alignment with the national geospatial infrastructure.

Senior Environmental Survey Specialist and GIS Lead **Feb 2025 – Present**

FUEL Professional Consulting | Client: Royal Commission for Jubail and Yanbu (RCJY), JCPDI | Jazan, Saudi Arabia

- Lead GIS for environmental baseline and planning studies supporting the development of JCPDI, an RCJY industrial city in Jazan.
- Designed the spatial database architecture for 300+ sampling stations and managed integration of satellite imagery, drone surveys, and field instrument data into a single project geodatabase.
- Delivered regulatory-grade geospatial products for RCJY submission: sediment trace-metal interpolation surfaces, supervised benthic habitat classification, and a marine environmental atlas.
- Processed Sentinel-2 multispectral imagery in Google Earth Engine for water quality mapping across the project area.
- Authored the JCPDI standards chapter on Digital Enablement and Smart Sustainability, defining normative requirements and KPI frameworks anchored to Saudi regulations.
- Resolve RCJY client review comments on the JCPDI Industrial and Municipal Waste Management Strategy reports; author data management plans and regulatory compliance documentation.

Postdoctoral Research Scientist **2023 – 2024**

Center for Satellite and Earth Science Research, George Mason University (NOAA and NASA funded) | Virginia, USA

- Contributed to NOAA- and NASA-funded satellite Earth observation research as a member of the center's research team, supporting project planning, milestones, and deliverables.
- Processed and integrated multi-source geospatial data (satellite imagery, airborne sensors, and in situ measurements) to agency data standards and interoperability requirements.
- Produced satellite-derived environmental monitoring datasets, technical reports, and briefings for federal program reviews.

GIS and Remote Sensing Consultant, Landslide Hazard Assessment **Jan 2018 – May 2018**

City of Los Angeles, Board of Public Works | Los Angeles, California, USA

- Built parcel-level landslide susceptibility maps integrating slope, lithology, drainage, land cover, and rainfall data to prioritize inspections and mitigation.
- Processed airborne LiDAR and aerial imagery into high-resolution DEMs and DEM-of-difference models to map scarps, runout zones, terrain deformation, and material loss.
- Published an ArcGIS Online web map and dashboard for the Emergency Operations Center and Public Works, supporting road closure, utility protection, and access decisions.
- Automated geoprocessing with ModelBuilder and Python; documented QA/QC procedures and metadata for auditable city outputs.

Graduate Research Scientist, Geospatial Intelligence and Remote Sensing **Aug 2015 – May 2023**

George Mason University, in collaboration with the U.S. Geological Survey (USGS) | Fairfax, Virginia, USA

- Delivered two USGS hyperspectral mapping projects in Colorado and Alaska: scoped requirements, led field data collection, processed airborne imagery, and delivered outputs to federal stakeholders on schedule.
- Developed five Python geospatial processing applications using PostgreSQL/PostGIS, Docker, and cloud storage (GCP, AWS S3), improving data pipeline efficiency by **60%**.
- Applied machine learning (CNN, SVM, Random Forest) to satellite image classification, improving accuracy by **50%**; processed terabyte-scale satellite archives on Google Earth Engine.
- Led geospatial tools and data workflow training for graduate students and industry partners.

Geological Engineer, Remote Sensing and Geospatial Analysis **Feb 2012 – Apr 2014**

Saudi Geological Survey (SGS) | Jeddah, Saudi Arabia

- Delivered geospatial analysis for national mineral exploration programs, improving resource estimation accuracy by **20%** and reducing data processing time by **25%**.
- Processed Landsat and ASTER multispectral imagery (band ratios, principal component analysis) to map hydrothermal alteration and lithology and delineate exploration targets.
- Compiled geological, geochemical, structural, and remote sensing layers into project geodatabases and produced GIS maps for exploration programs.
- Conducted field surveys to ground-truth remote sensing interpretations, collecting GPS-located observations and samples.
- Led cross-functional R&D on new data collection and processing techniques, improving geological prediction accuracy by **25%**.
- Trained junior geologists on remote sensing and exploration technologies, raising team efficiency by **15%**.
- Prepared technical reports and implemented spatial data standards with cross-functional government and industry teams.

Process Engineer **Mar 2006 – Feb 2012**

Riyadh Cement Company | Riyadh, Saudi Arabia

- Process engineer on a **$280M, 5,000 TPD** clinker cement plant project delivered with Sinoma International as EPC contractor, working with the contractor, equipment suppliers, and plant teams.
- Supported the **$40.7M** Muzahmiyah plant expansion with Fives FCB that raised clinker production from 700 to 1,000 TPD.
- Monitored kiln, raw mill, and cement mill operations; performed heat and mass balances and troubleshot process deviations with operations and maintenance teams.
- Analyzed kiln and process data to reduce energy consumption by **20%** and increase production capacity by **15%**; improved Overall Equipment Effectiveness (OEE) by **15%** through statistical tracking.
- Worked with cross-functional teams to implement advanced process control and applied FMEA to prioritize reliability improvements on critical equipment.
- Presented performance metrics and ROI projections to senior management to secure investment in efficiency projects.

SELECTED PROJECTS

- **BIM in ArcGIS (GeoBIM):** Author of BIM in ArcGIS: A Student Cookbook (2nd ed.), teaching BIM model integration in ArcGIS Pro.
- **Saudi Data Center Water-Stress Map:** Water-stress map of Saudi data center sites (WRI Aqueduct 4.0, Leaflet, ArcGIS Online); IAEE MENA 2026 abstract.
- **Yemen MASAM:** Mine-contamination susceptibility model built in Google Earth Engine.
- **Wadi Ibrahim, Makkah:** 40-year multi-temporal satellite land-use change analysis for a book in progress on the wadi.
- **300 Days Mapping Challenge:** Self-directed series of daily interactive geospatial maps of the Jazan region.

TECHNICAL SKILLS

Enterprise GIS: ArcGIS Pro, ArcGIS Enterprise, ArcGIS Online, ArcSDE, ArcGIS Dashboards, Field Maps, Survey123, QGIS

Data Management: PostgreSQL/PostGIS, Oracle, MySQL; enterprise geodatabase design; schema and data model management; spatial ETL; metadata (ISO 19115); QA/QC

Integration: BIM/CAD to GIS (Revit, Civil 3D, ArcGIS Pro BIM workflows), FME, GDAL/OGR; drone, LiDAR, GPS survey, and satellite data; coordinate system and datum transformations

Spatial Analysis: Network and routing analysis, suitability and multi-criteria analysis, spatial interpolation, change detection, terrain and LiDAR analysis, image classification

Programming: Python (ArcPy, ArcGIS API for Python, GeoPandas, Rasterio, scikit-learn, TensorFlow), SQL, R, JavaScript (Leaflet, GEE API), Git/GitHub, Docker

Remote Sensing: Hyperspectral, multispectral, SAR, LiDAR; ENVI, ERDAS IMAGINE, SNAP, Google Earth Engine; Sentinel-2, Landsat, WorldView

Cloud, Web, and Standards: GCP, AWS (S3, EC2), Azure; Leaflet, OpenLayers, Mapbox; GEOSA, OGC, ISO 19115

EDUCATION

PhD, Earth Systems and Geoinformation Sciences, *George Mason University, Virginia, USA*
Dissertation: Integrating Nonlinear Kernel-Based Detector Algorithms and Minimum Noise Fraction for Advanced Oil-Spill Detection with Hyperspectral Imagery

MS, Engineering Management, *Gannon University, Pennsylvania, USA*

BS, Geological Engineering and Environmental Science, *King Abdulaziz University, Jeddah, Saudi Arabia*

PROFESSIONAL CERTIFICATIONS

Certification	Issuing Organization	Year
Multisensor Data Fusion for Object Detection, Classification and Identification	SPIE	2019
Hyperspectral Data Analysis	Harris Corporation	2018
Hyperspectral Data Processing	L3Harris Technologies	2017
Fundamentals of SAR Signal Processing	Georgia Institute of Technology	2017
SAR Image Formation Processing	Georgia Institute of Technology	2017

PROFESSIONAL AFFILIATIONS

- American Society for Photogrammetry and Remote Sensing (ASPRS)
- United States Geospatial Intelligence Foundation (USGIF)
- International Society for Optics and Photonics (SPIE)
- IEEE Geoscience and Remote Sensing Society
- The Geological Society of America (GSA)